NN-H-101F
March 12, 1973
SUPERSEDING
Fed. Spec. NN-H-101E
May 8, 1962

FEDERAL SPECIFICATION

HANDLES, MOP (WET)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wood and metal mop handles and three basic types of attached fixtures primarily intended for use with wet-cleaning mopheads.

1.2 Classification.

1.2.1 Types, classes and sizes. Mop handles and attachments covered by this specification shall be of the following types, classes, and sizes (length), as specified in the invitation for bids (see figure 1 and 6.3):

Type I - Screw type.
  Classes:
  1 - Wood handles.
    Size A - 54 inches.
    Size B - 60 inches.
    Size C - 72 inches.
  2 - Steel or aluminum handles with plastic sleeve.
    Size A - 54 inches.
    Size B - 60 inches.
    Size C - 72 inches.

Type II - Spring-lever type.
  Class 1 - Wood handles.
    Size A - 54 inches.
    Size B - 60 inches.

Type III - Clip type.
  Class 2 - Steel or aluminum handle with plastic sleeve.
    Size A - 54 inches.
    Size B - 60 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-C-320 - Chromium Plating (Electrodeposited).
QQ-N-290 - Nickel Plating (Electrodeposited).
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements for.
CCC-C-467 - Cloth, Burlap, Jute (Or Kenaf).
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Cornered.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 406 - Plastics, Method of Testing.
Fed. Test Method Std. No. 601 - Rubber; Sampling and Testing.

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials (general). The mop handle shall consist of wood, plastic covered steel or plastic covered aluminum alloy handle to which are attached metal fixtures for securely holding a mop.

3.1.1 Wood handle. The wood for the handles shall be of straight-grained ash, beech, birch, elm, cherry, hickory, maple, pecan, red gum, red oak, white oak, ramin (Gonystylus Spp.), Douglas fir, or Southern yellow pine. The wood shall be free from cup, knots, shakes, wane, worm holes, heart center, rot, dote, brash, and cross grain and spiral grain with a slope greater than 1-inch in 15 inches. Moisture content shall be not more than 12-percent when tested as specified in 4.4.4.

3.1.2 Steel or aluminum alloy handle. At the option of the contractor, the tubing for the handles may be steel or aluminum alloy. Steel tubing shall be AISI 1020 to 1030 strip steel of not less than 0.029-inch wall thickness. Aluminum alloy tubing shall be aluminum-manganese, 3000 series of not less than 0.057-inch wall thickness and the temper shall be strain hardened and partially annealed. The tubing shall be of a seamless of welded construction in one section measuring not less than 7/8-inch outside diameter. The steel or aluminum alloy tubing shall be covered with a colored plastic sleeve.

3.1.3 Steel, metal fixture. The steel for the metal fixture (mop holder) shall be of the type and class used commercially unless otherwise specified in this specification.

3.1.4 Vinyl plastic for metal handle. The plastic sleeve shall be polyvinylchloride of 90 hardness when tested as specified in 4.4.3. The wall thickness shall be not less than 0.025-inch and shall be shrunk tightly on the tubing. The plastic sleeve shall be unaffected by a high-low temperature test as specified in 4.4.1. The sleeve shall withstand a voltage of not less than 6,000 RMS, 60 C/S when tested as specified in 4.4.2. The finished handle shall have a molded dome-shaped vinyl cap of the same material and color as the sleeve, or a rubber or vinyl plastic handle grip securely cemented to the hand end of the handle forming a closure. The mop end shall be plugged with vinyl plastic to prevent water entering the handle.

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3.2 Design. Mop handles shall be of the general designs shown in figure 1. The illustrations are included for convenience in identifying each mop handle type and are not to preclude designs otherwise meeting the requirements of this specification.

3.2.1 Type I, screw type. Screw type mop handles shall consist essentially of a steel or aluminum handle with plastic sleeve and metal fixture, or a wood handle and metal fixture for holding a wet-cleaning mop between the head and bail by tightening a wingnut over a screw pin. (See figure 1, type I.)

3.2.2 Type II, spring type. Spring type mop handle shall consist essentially of a wood handle and metal fixture for holding a wet-cleaning mop between the head and bail by means of a spring and lever. (See figure 1, type II.)

3.2.3 Type III, clip type. Clip type handles shall consist essentially of a steel or aluminum handle with plastic sleeve and metal fixture for holding a wet-cleaning mop by means of a detachable bail (spring yoke). (See figure 1, type III.)

3.3 Construction.

3.3.1 Type I. The metal fixture for holding a wet-cleaning mop consists essentially of a bail, ferrule, head, screw pin, and wingnut. The length of the fixture from the top of the ferrule to the base of the bail shall be 10-1/2 inches ± 1/4-inch. The handle shall be securely fastened to the fixture.

3.3.1.1 Bail and ferrule. The bail shall be made of steel wire not less than 0.246 ± 0.01-inch in diameter, and shall be securely fastened to the wood or metal handle by means of a steel ferrule. The ends of the bail shall enter the ferrule for not less than 2-1/4 inches and abut against the end of the handle and shall be securely fastened in such a manner that it will not be possible to separate the ends of the bail from the ferrule. The ferrule shall consist of two pieces of cold-rolled steel not less than 0.071-inch in thickness, die-formed to closely fit the bail wire and the handle. The identical halves of the ferrule shall be riveted tightly together at the base of the ferrule with two 5/32-inch diameter steel rivets 1/4-inch long, and riveted tightly to the handle, inserted not less than 1-7/8 inches, with one 3/16-inch diameter rivet 1-3/8 inches long. The inside dimensions of the bail shall be 6-7/16 inches ± 1/8-inch provided the bail moves freely in the head without coming out of the lips at the end.

3.3.1.2 Head. The head shall consist of two identical pieces of die-formed cold-rolled steel of not less than 0.045-inch in thickness. They shall be spotwelded together at not less than six equidistant points. An opening shall be provided in the head to receive the screw pin. There shall be two continuous ears running the length of the bottom of the head to firmly grip the mop, and ears at both ends in order to guide the head along the bail. With the head raised to a maximum, there shall be an opening of at least 1-1/4 inches between the base of the bail and bottom of the head to provide for the insertion of the mop.

3.3.1.3 Screw pin. The screw pin shall be firmly locked in the ferrule and threaded to 16 UNC threads per inch to accommodate a hot forged low wingnut with a minimum of four complete 16 UNC threads.

3.3.2 Type II, spring-lever type. The metal fixture for holding the mop shall consist essentially of a bail, head, spring, and lever.

3.3.2.1 Bail. The bail shall be made of steel wire 0.177-inch ± 0.015-inch in diameter, securely fastened to the lever. The inside dimension of the bail shall be 6-7/16 inches ± 1/8-inch in width. When the lever is fully raised, there shall be an opening of at least 1-1/4 inches between the end of the bail and the head for the insertion of the mop.

3.3.2.2 Head. The head shall be made of malleable iron or pressed steel. If pressed steel is used it shall be not less than 0.0265-inch in thickness when used double thickness and 0.0306-inch in thickness when used single thickness. The head shall be fastened to the handle by driving a 12-gage-steel wire pin through the head until it protrudes on the opposite side. The steel pin shall be approximately 1-1/4 inches long, shall have a rounded head, and shall be peened on the opposite end. Alternatively the head may be securely riveted to the wooded handle.

3.3.2.3 Spring. The spring shall be made of spring-steel wire 0.135 ± 0.008-inch in diameter. It shall be so constructed that sufficient pressure will be produced to maintain the mop in a secure position.

3.3.2.4 Lever. The lever shall be made of pressed steel. Steel 0.049 ± 0.004-inch in thickness shall be used if the lever is embossed for added strength; steel 0.061 ± 0.004-inch in thickness shall be used if the lever is plain. Where no collar is provided, the lever shall be securely attached to the lower turn of the spring so as not to become detached.

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3.3.3 Type III, clip type. Clip type metal fixtures consist essentially of a bail (spring yoke) and a head. The length of the fixture shall be 4-3/4 inches ± 1/4 inch.

3.3.3.1 Bail (spring yoke). The bail shall be made of tempered spring-steel wire 0.162 ± 0.015 inch in diameter, shaped so as to be firmly attached and removable from the head.

3.3.3.2 Head. The head shall consist of two identical pieces of die-formed cold-rolled steel of not less than 0.0359-inch in thickness spotwelded together at not less than six equidistant points. There shall be two rivets measuring not less than 3/8-inch in length and not less than 3/16-inch in diameter, each corner of the head having one semi-tubular rivet protruding through both pieces of the head, leaving a track not less than 3/16-inch wide in which the bail (spring yoke) can rest. Both protruding rivets must be spun flat and smooth. The butt end of the head shall be flared to form a groove 1/4 ± 1/16-inch deep in which the string mop shall remain stationary without being abraded when the bail (spring yoke) is secured. Two metal tabs (hooks) 3/4 ± 1/8-inch shall be provided, one on each side of the head 3-1/2 inches ± 1/8-inch along the handle in which to lock the bail (spring yoke). Metal tabs shall be an extension of the two pieces forming the head. The metal fixture shall be securely fastened to the handle with rivets.

3.4 Dimensions.

3.4.1 Wood handles. Wood handles shall be round with a minimum diameter of 1-1/8 inches plus 1/8-inch tolerance. The length shall be as specified ± 1/2-inch tolerance.

3.4.2 Steel or aluminum alloy handles. Steel or aluminum alloy handles shall have a minimum outside diameter of 7/8-inch before covering with the plastic sleeve. The length shall be as specified ± 1/2-inch tolerance.

3.4.3 Metal fixtures (heads). The metal heads shall conform to the dimensions as specified for each type.

3.5 Finish.

3.5.1 Hardwood handles. Hardwood handles shall be uniform in diameter and rounded at the hand end. The handles shall be sanded smooth and coated with a clear lacquer or varnish resulting in a uniform glossy surface for all exposed parts. Douglas fir and Southern yellow pine handles shall be sanded smooth and finished with a heavy coating of hard vinyl plastic impervious to strong detergents commonly used for extremely dirty floors.

3.5.2 Steel or aluminum alloy handles. Handles shall be finished with the plastic sleeve (see 3.1.4) in accordance with the best commercial practice.

3.5.3 Metal components (except handles). All metal components shall be made of corrosion-resistant materials or shall be made corrosion-resistant with either zinc plating conforming to type I, class 2 of QQ-Z-325 or cadmium plating conforming to type I, class 2 of QQ-P-416 or chromium plating conforming to type I or II, class 1 of QQ-C-320 or nickel plating conforming to type I, II, or III, class 1 of QQ-N-290.

3.6 Workmanship. Metal portions of the handles shall be well secured and shall be free from burrs or roughened spots. The wood or metal handles shall be free from defects which affect their appearance or which might impair their serviceability. The handles (mop) shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the levels set by the applicable acceptable quality level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the examination and testing set forth in the specification where such are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise specified.

4.2.1 Inspection of components and materials. In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with the specified requirements or referenced subsidiary specifications and standards or if none, in accordance with this specification.

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4.2.1.1 Dimensional examination. Each component or material which cannot be measured after assembly shall be examined in lots, by sampling, for compliance with this specification and applicable specifications. Any deviation therefrom shall constitute a defect. The acceptance level shall be level I, AQL 2.5. If the number of defective units exceeds the acceptance number specified in MIL-STD-105, the entire lot of handles shall be considered as having failed to meet the requirements of the specification.

4.2.1.2 Testing of components. Testing shall be performed on each sample unit fixture (all types) for thickness of zinc or cadmium coating, as applicable, on metal fixtures as specified in 3.5.3. The inspection level shall be level S-3, AQL 1.5 percent defective.

4.3 End item examination and testing.

4.3.1 Lot. For the purpose of examination and testing of the end item, a lot shall consist of all handles (mop) of one type and class offered for delivery at one time.

4.3.2 Visual and dimensional examination. The defects found during examination shall be classified in accordance with the list shown in table I. The sample unit for this examination shall be one complete mop handle. The lot shall be unacceptable if the unit fails to meet any requirement classified in accordance with the acceptable quality levels (AQL's).

4.3.2.1 Acceptable quality levels (AQL's). The acceptable quality levels expressed in percent defective shall be as follows for the defects listed in table II.

	AQL
Major	1.5
Minor	4.0

TABLE I. Visual and dimensional defects

Examine	Defect	Classification	
		Major	Minor
Handles	Not hardwood, Southern yellow pine, or Douglas fir.		X
	Any decay or brashness (appearing as soft, spongy, or punky wood).	X	
	Slope of grain steeper than 1-inch in 15 inches or dip grain more than one-half the diameter.		X
	Not close grain.		X
	Any knot hole or loose knot.	X	
	Crook (warp or bow) more than 3/8 of an inch per full length of handle (measure at point of greatest distance from straight line).	X	
	Any tight or firmly encased knot 1/8 of an inch or more in diameter.		X
	Any season check, fracture, shake, or split open or visibly concealed with filler.	X	
	Any wane (bark or lack of wood).	X	
	Pronounced streak or stain obscuring grain of woods.		X
	Any wormhole, larva, channel, or pocket open or visibly concealed with filler.		X
	Loose, raised, or torn grain producing a rough surface along upper two-thirds of handle.	X	
	Dimension not as specified.	X	

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TABLE I. Visual and dimensional defects (con.)

Examine	Defect	Classification	
		Major	Minor
Finish of handles	Hardwood not finished with lacquer, varnish or other suitable coating. Douglas fir and Southern yellow pine not vinyl coated.	X	
	No plastic sleeve on metal handle (class 2 only).	X	
	Dirt or foreign matter imbedded in finish.		X
	Wet or tacky, uncoated areas.	X	
	Not smoothly sanded, not rounded at hand end.		X
	Out of round (any flat area or lack of uniform diameter throughout length).		X
	Consecutive machine skip, tool mark, or gouge along upper two-thirds of handle.	X	
	Occasional machine skip, tool mark, or gouge along upper two-thirds of handle.		X
Handle fixture	Any part missing, and failure to meet specified type and size.	X	
	Not electroplated zinc coating or cadmium plated.	X	
	Area of no coating; blistered; evidence of rust.	X	
	Dimensions not as specified.	X	
	Damaged or malformed, affecting serviceability.	X	
	Any sharp burrs or metal slivers that may cause injury	X	
	Welds broken, splattered not removed, burned through, less than specified number of welds.	X	
	Any part that will not operate or require abnormal force to operate.	X	
	Fixtures do not firmly grip mop.	X	
	Locking device does not function properly.	X	
	Any component not fabricated of the specified material (unless otherwise indicated).	X	

4.3.3 Examination of preparation for delivery. An examination shall be made to determine whether the markings, material, workmanship and contents comply with the requirements of section 5. The sample unit shall be one shipping container fully packed, selected just prior to closing operations. The closed shipping containers shall be examined for closure defects. The inspection level shall be S-2 and the AQL shall be 4.0 expressed as defects per hundred units.

TABLE II. Examination for delivery

Examine	Defects
Marking (interior and exterior container)	Incorrect; incomplete; illegible; omitted; of improper size; location, sequence, or method of application.
Materials	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of flaps of containers, loose or inadequate sealing, strapping, or stapling, when required.

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TABLE II. Examination for delivery (con.)

Examine	Defects
Contents	Number of unit containers in the shipping container more or less than required.
Containers	Not as specified; not close-fitting when required for level A or B.

4.3.4 Testing of end item. Handles shall be tested as specified in 4.1 through 4.4.4 for conformance with the applicable requirements of this specification. The sample unit shall be two handles. The inspection level shall be S3 and the acceptable quality level (AQL) 1.5 percent defective. If any handle in the sample unit fails to meet any requirement for which tests are made, the sample unit shall be considered defective. If the number of defective sample units in the sample for test exceeds the acceptance number specified in MIL-STD-105, the entire lot of handles shall be considered defective.

4.4 Tests.

4.4.1 High-low temperature, plastic. A plastic-sleeved metal handle or any portion thereof required to make the test shall be subjected to a high-low temperature in a circulating air cabinet or oven at - 40°F, as described in No. 6011/Fed. Test Method Std. No. 406.

4.4.2 Dielectric strength, plastic. A plastic sleeved metal handle or any portion thereof required to make the test shall be subjected to 6,000 volts for 1-minute in accordance with the test described in No. 4031/Fed. Test Method Std. No. 406. Test shall be conducted at room temperature.

4.4.3 Hardness, plastic. A plastic sleeved metal handle or any portion thereof required to make the test shall be subjected to the hardness test as described in No. 3021/Fed. Test Method Std. No. 601; using the type A Shore Durometer.

4.4.4 Moisture content of wood. The moisture content of wood handles shall be determined at a distance of two feet from the end of the handle and shall be measured by an electrical moisture meter or other suitable method.

4.4.5 Thickness of cadmium or zinc. The thickness of cadmium or zinc finish imparted to the fixture and component parts shall be determined in accordance with QQ-P-416 and QQ-Z-325 respectively.

5. PREPARATION FOR DELIVERY

5.1 Packaging. (not required.)

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

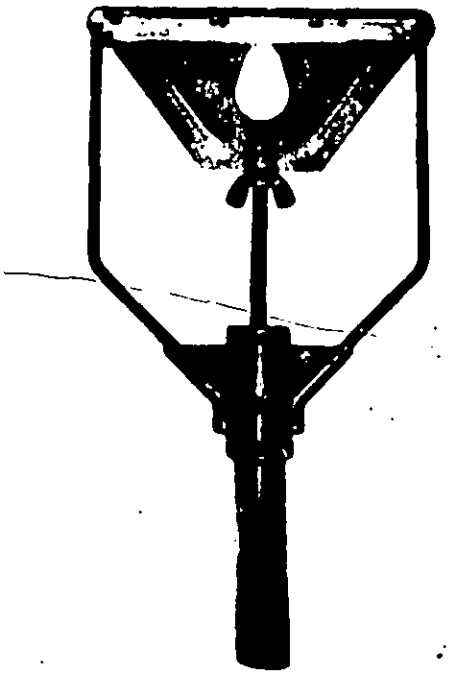
5.2.1 Level A. *Thirty-six wood handles or 24 steel handles with attachments shall be packed in a* close-fitting box conforming to any of the following overseas type container, PPP-B-585, class 3; PPP-B-601, overseas type, PPP-B-621, class 2; or PPP-B-636, class weather-resistant. One-half of the heads shall be placed at each end of the box. When fiberboard boxes are used, both inside ends of the boxes shall be provided with scored and slotted trays having 3-inch wide flanges all around, fabricated of the same material as the fiberboard box. The boxes shall be closed in conformance to the appendix of the appropriate box specification.

5.2.2 Level B. Thirty-six wood handles or 24 steel handles shall be packed in a close-fitting fiberboard box conforming to PPP-B-636, class domestic. The heads of the handles shall be alternately located at opposite ends of the box. The ends of the box shall be provided with scored and slotted trays having 3-inch wide flanges all around, fabricated of the same material as the box. The boxes shall be closed in accordance with the appendix to PPP-B-636.

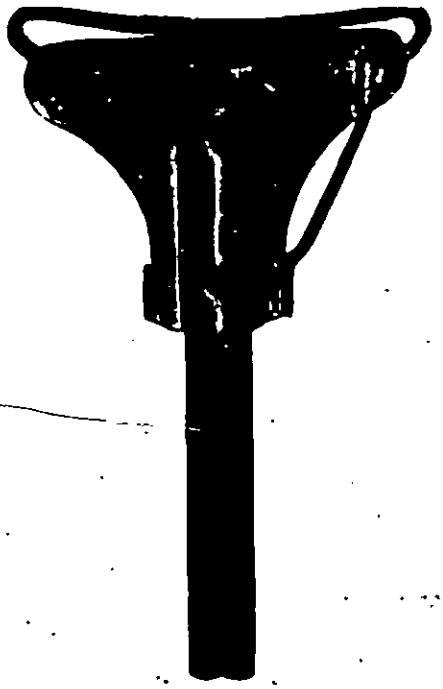
5.2.3 Level C. The wood or steel handles shall be packed so as to insure arrival at destination in a satisfactory condition in containers that comply with the Uniform Freight Classification Rules or the National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any marking required by the procurement documents, the shipping containers shall be marked in accordance with Fed. Std. No. 123 for civilian agencies or MIL-STD-129 for Military activities, as applicable.

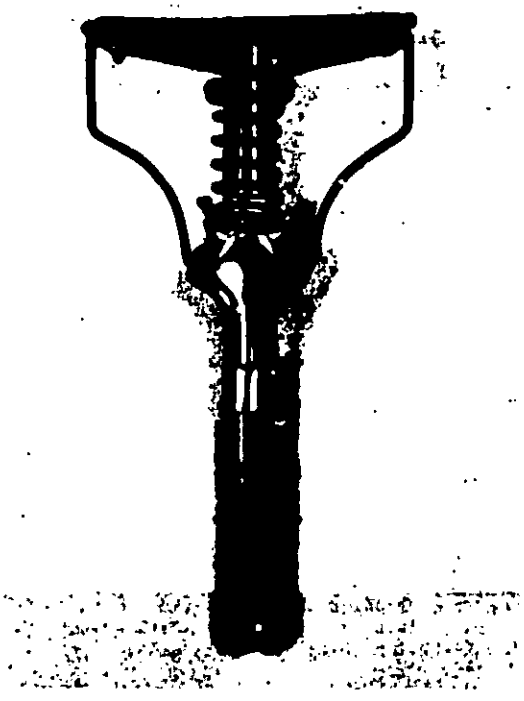
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Type I - Screw type



Type III - Clip type



Type II - Spring-lever type

FIGURE 1 - Mop handles

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6. NOTES

6.1 Intended use. This specification is for mop holders to be used with wet-cleaning mopheads.

6.2 Standard pack. The Standard Pack requirements in 5.2.1 are intended for use in procurement of Stores Stock replenishments. *Procuring officers should use the Standard Pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or intermediate holding point for temporary storage, subsequent issue, or shipment to eventual user.*

6.3 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Types, classes, and sizes, (see 1.2.1).
(b) Whether mop handles shall be packed for level A, level B, or level C (see 5.2).
(c) Marking (see 5.3).

6.4 Classification changes. This revision deletes clip type under type I and adds the type III clip (steel or aluminum handle - plastic sleeve).

MILITARY INTEREST

User activity:

Army - GL

DOD has waived coordination on revisions and amendments to this specification until further notice.

Civil Agency Coordination Activity:

GSA-FSS

Preparing Activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

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